

Three failure classes, read across the four units of analysis

Classification and exemplars transcribed from E1-DATASET-v1.0.md (frozen 2026-09-29, corrected 2026-09-30), section 'The three counterexample classes', and from the cross-package findings in E1-AGGREGATE.md. A cell is a structural property of the class definition, not a count: it records whether the class is detectable at that unit at all.

failure class and corpus exemplar	U1	U2	U3	U4	mechanism the method must be able to falsify
Circular validation isac-jasc-ieee	NO	YES	NO	YES	Artifact reproduces a property derived from the assumption it purports to validate
Contradictory verification ems-ieee-package	YES	YES	NO	YES	Verifier executes and its result contradicts the claim it is supplied to verify
Semantic correspondence failure quantum-k-sat-ieee-package	NO	YES	YES	YES	Artifact exists and runs but implements a different mechanism from the one it names

Read the four cell columns down the three rows. Class 1 is invisible at U1, U2 and U3 at once: its artifact reproduces to 1.8e-12, the most reproducible artifact in the set, and no artifact contains a different value for the asserted proposition. It is detectable only at U4, and only because the assertion's own construction is the defect.

Classes 2 and 3 are both invisible at U2, because reproduction is not what is wrong with them: a verifier that runs and disagrees, and an artifact that runs and implements something else, both reproduce perfectly. The two failure classes a reproduction check cannot see are precisely the two the earlier unit set did not have.